

End-to-End E-Commerce Sales & Customer Analytics Dashboard

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Tools: Python | SQL | Power BI
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1. Introduction

This project presents an end-to-end data analytics solution developed to analyse e-commerce sales performance, customer behaviour, and operational efficiency. The objective is to transform raw transactional data into actionable insights using Python, SQL, and Power BI.

2. Objectives

- Analyse revenue trends
- Understand customer behaviour
- Evaluate delivery performance
- Identify factors affecting customer satisfaction
- Build an interactive dashboard for decision-making

3. Tools & Technologies

- Python (Pandas, NumPy)
- SQL (PostgreSQL)
- Power BI
- Excel

4. Dataset Description

The dataset used in this project is based on the Olist Brazilian E-commerce dataset, a publicly available dataset providing detailed transactional data from a real-world online marketplace. It includes customers, orders, products, payments, reviews, and sellers. These tables were merged to create a unified dataset enabling comprehensive analysis of sales performance, customer behaviour, delivery efficiency, and payment trends.

5. Data Cleaning & Preparation

- Handled missing values
- Removed duplicate records
- Converted date formats
- Merged multiple datasets
- Created features such as delivery time and delivery status

6. Data Analysis

SQL was used to extract key business metrics including total revenue, order volume, and customer counts. Additional analysis included monthly revenue trends, product category performance, and payment method distribution, forming the basis for dashboard KPIs and visualisations.

7. Dashboard Overview

The Power BI dashboard is divided into two sections: Business Overview and Operations & Customer Insights. It includes KPIs, trends, geographic distribution, and operational performance metrics.



Figure 1: Business Overview Dashboard



Figure 2: Operations & Customer Insights Dashboard

8. Key Insights

- Late deliveries have a significant negative impact on customer satisfaction
- Most deliveries are completed within 5–10 days
- Credit card payments contribute the highest share of revenue
- A small number of product categories account for a large portion of total sales

9. Business Recommendations

- Improve delivery timelines to enhance customer satisfaction and retention
- Focus on high-performing product categories to maximise revenue
- Optimise payment processes to support preferred payment methods
- Continuously monitor performance using dashboards

10. Limitations

This analysis is based on historical data and does not include real-time updates or external factors such as marketing campaigns or seasonal effects.

11. Conclusion

This project demonstrates end-to-end data analytics capability and highlights how insights can support better business decision-making.